

**Time:** 5 minutes**Outcome:** Identify Efficient Quantity & Allocation.**Difficulty:** ●●○ Intermediate

Efficient Quantity & Allocation



THE CORE IDEA

The efficient quantity includes every unit whose buyer value is at least as large as its opportunity cost. Efficiency pairs the highest-value buyers with the lowest-cost sellers and continues exchange while the next buyer's willingness to pay covers the next seller's cost. Producing beyond that point destroys surplus, while stopping before it leaves gains unrealized.



HOW TO RECOGNIZE IT

- Order potential trades from the highest buyer value and the lowest seller cost.
- Complete every trade for which willingness to pay is at least as large as opportunity cost.
- Stop when the next seller's cost exceeds the next buyer's value.
- An efficient allocation sends the good to the highest-value buyers and production to the lowest-cost sellers.



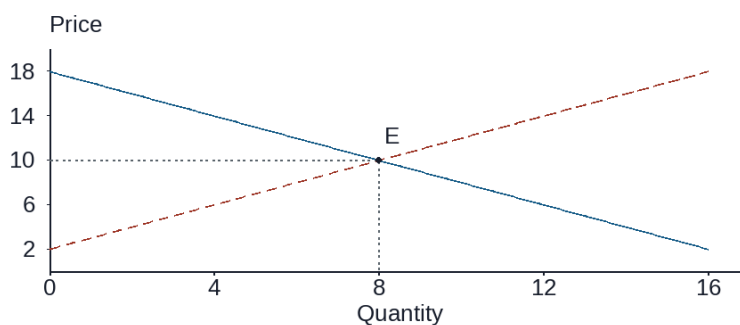
WATCH OUT

More output is not always more efficient. Stop when the next unit costs more to produce than its buyer is willing to pay.



WORKED EXAMPLE: FINDING THE EFFICIENT QUANTITY

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Demand and supply intersect at 8 units and \$10. At that quantity, the marginal buyer's value equals the marginal seller's cost. Units below 8 create positive gains from trade; units above 8 cost more than buyers value them. The efficient allocation therefore produces 8 units.



CHECK YOURSELF

Five buyers value one pass each at \$27, \$23, \$18, \$13, and \$8. Five sellers have costs of \$4, \$10, \$15, \$21, and \$25. How many trades create nonnegative gains, and why should the next trade not occur?

**READY?**

Return to the game and master this concept.